

Avi Walia

647-637-8025 | avi.walia@mail.utoronto.ca | [linkedin.com/in/avi-walia](https://www.linkedin.com/in/avi-walia) | github.com/Avi04w

EDUCATION

University of Toronto

Sep 2022 – Apr 2026

Honours Bachelor of Computer Science, Specialization in Data Science

Toronto, ON

Relevant Courses: Data Science, Machine Learning, Neural Networks and Deep Learning, Databases, Data Visualization, Data Analysis, Algorithms, Computational Linguistics, Software Engineering

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, JavaScript, C, HTML/CSS

Libraries/Frameworks: Pandas, NumPy, Scikit-Learn, PyTorch, Matplotlib, React.js, Flask, JUnit

Developer Tools: AWS, Git, Power BI, Tableau, Jupyter, Google Colab, Maven, Docker

Databases: Oracle DBMS, Google Firestore, PostgreSQL, MongoDB, DynamoDB

Skills: Data Science, Software Engineering, ETL, Cloud Computing, Web Development

EXPERIENCE

Software Development Engineer I

June 2025 – Present

Amazon – Privacy Engineering

Toronto, ON

Data Science Student Researcher

Sep 2025 – May 2026

Centre for Management of Technology & Entrepreneurship - University of Toronto

Toronto, ON

- Collaborated with an industrial partner to design a machine learning framework that informs strategic inventory decisions through data-driven product hierarchy classification.
- Analyzed 9M+ transaction records across 160K+ products to identify co-purchase dependencies and demand hierarchies, improving stocking strategy and minimizing missed sales opportunities.

Software Development Engineer Intern

May – Aug 2025

Amazon – Customer Service Operating System

Toronto, ON

- Reduced a 60 hour build cycle to under 1 hour by owning end-to-end development of an automation tool, significantly accelerating deployment of secure device images for Customer Service Agents.
- Designed and implemented the solution using AWS services including Lambda, S3, DynamoDB, ECS, EC2, and SQS, ensuring scalability, reliability, and maintainability.

Full Stack Developer Intern

May – Aug 2023 and May – Aug 2024

Third Eye Solutions Inc.

Mississauga, ON

- Facilitated and carried out application upgrade from Java 8 to 17, improving performance and security.
- Used Oracle SQL Developer to write queries to maintain Customer Schemas and update Data Models.

PROJECTS

Toronto Subway Delay Analysis | *R, Regression, GAM, Random Forest, XGBoost, Data Visualization*

- Developed predictive models analyzing 6,500+ TTC subway delays to uncover operational bottlenecks.
- Engineered a structured dataset by filtering invalid and multi-station entries, integrating delay code mappings, and aligning all records to unified time indices.

Twitter Trending Topics Data Analysis | *Python, pandas, NumPy, scikit-learn, NLTK*

- Conducted trend analysis on recent tweets mentioning "Toronto" using Latent Dirichlet Allocation.
- Utilized Python libraries such as Tweepy and NLTK to extract, clean, and preprocess social media data.

EXTRA-CURRICULARS

- Vertige Investment Group - Project Lead of Algorithmic Trading
- Athletics - Intramural Sports Teams - Basketball and Volleyball